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ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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Cloud based intelligent pharmaceutical supply chain monitoring and counterfeit drug detection system

CSA1506

CLOUD COMPUTING AND BIG DATA ANALYSIS

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INTRODUCTION



- Pharmaceutical supply chains involve manufacturers, distributors, wholesalers, and retailers.
- Manual tracking can cause delays, data errors, and poor visibility.
- Counterfeit and expired medicines can create serious risks.
- Cloud computing enables centralized and real-time data management.
- The proposed system combines cloud monitoring, supply-chain tracking, and counterfeit detection.



PROBLEM STATEMENT



- Difficulty in tracking medicines across multiple supply-chain stages.
- Lack of centralized medicine information.
- Counterfeit medicines may enter the supply chain.
- Expired or unauthorized medicines may reach retailers.
- Manual monitoring makes detection and reporting difficult.



OBJECTIVES



- Register medicines and maintain their details securely.
- Track medicines from manufacturer to retailer.
- Verify medicines using **QR codes/barcodes**.
- Detect counterfeit, duplicate, expired, or unauthorized medicines.
- Provide real-time dashboards and alerts.
- Generate useful reports for supply-chain monitoring



PROPOSED SYSTEM



- A cloud-based platform for pharmaceutical supply-chain management.
- Stores medicine and transaction information in a centralized database.
- Tracks movement through every supply-chain stage.
- Uses QR/barcode verification for authenticity checking.
- Provides dashboards, alerts, and reports.
- Improves transparency and monitoring.

LITERATURE REVIEW

AUTHOR & YEAR	FOCUS / METHODS	KEY FINDINGS
Ghasemshirazi et al. (2025)	Zero Trust Architecture	Improves security, access control, and supply-chain resilience.
Dutta & Priya G. (2026)	Blockchain + IPFS + smart contracts	Enables secure, transparent drug tracking and verification.
Mole & Shaji (2026)	Blockchain + IoT + Hyperledger Fabric	Supports decentralized traceability and condition monitoring.
Heikamp et al. (2026)	RFID traceability security analysis	Reveals vulnerabilities and the need for stronger security.
Dubey (2026)	Blockchain-based Indian pharma framework	Enhances drug traceability, transparency, and counterfeit prevention.

EXISTING VS PROPOSED SYSTEM

EXISTING SYSTEM	PROPOSED SYSTEM
1. Manual record management is used for maintaining medicine and transaction details.	1. A cloud-based platform is used for pharmaceutical supply-chain management.
2. Drug verification mainly depends on basic barcode/QR-code checking.	2. Medicine authenticity is verified using QR/barcode scanning.
3. Supply-chain monitoring is performed periodically rather than continuously.	3. Medicine movement is tracked through every supply-chain stage.
4. Different stakeholders may maintain separate records, making data sharing difficult.	4. Medicine and transaction information is stored in a centralized database.
5. Counterfeit or expired medicines may be detected late during manual inspections.	5. Dashboards, alerts, and reports improve transparency, monitoring, and faster detection.



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TECHNOLOGY USED





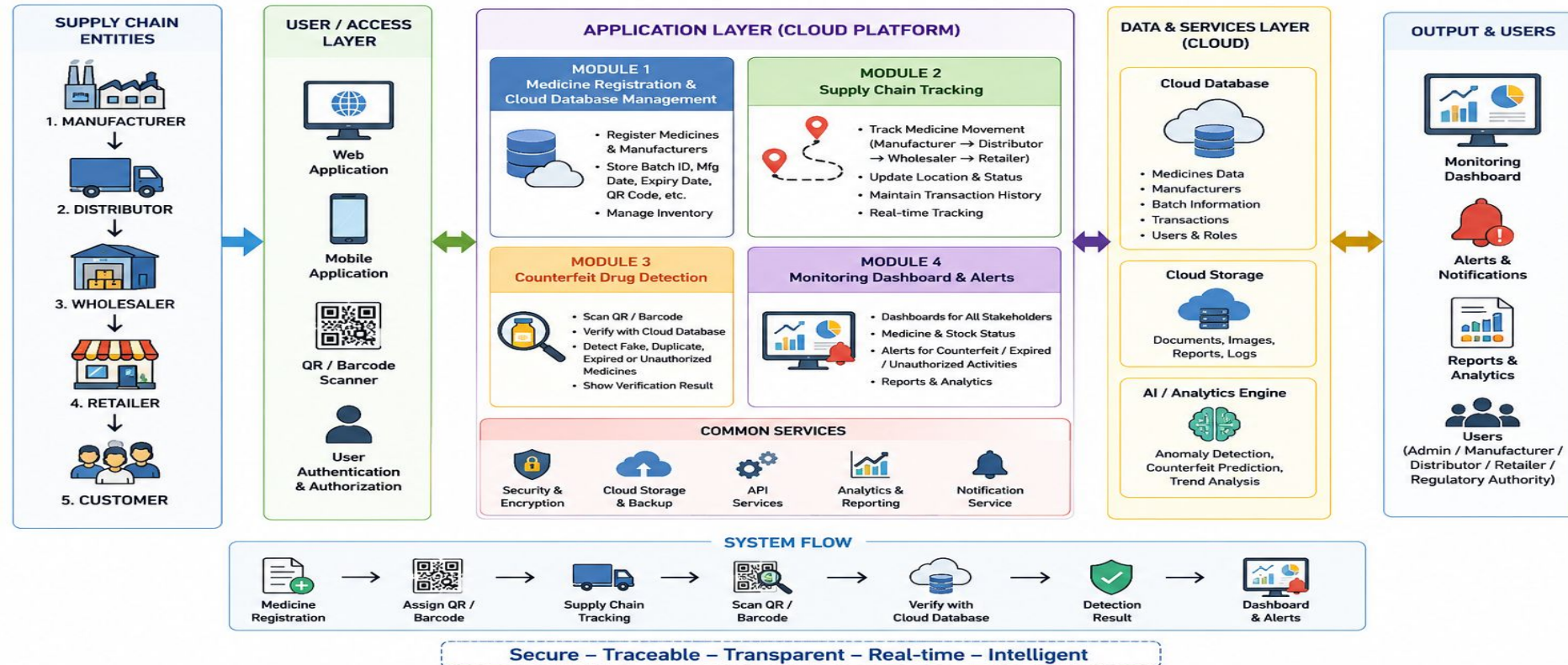
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SYSTEM ARCHITECTURE

SYSTEM ARCHITECTURE

Cloud-Based Intelligent Pharmaceutical Supply Chain Monitoring and Counterfeit Drug Detection System



Module 1: Medicine Registration & Cloud Database Management

- Registers manufacturers and pharmaceutical products.
- Stores Batch ID, Medicine Name, Manufacturer, Manufacturing Date, Expiry Date.
- Generates or records unique QR/barcode information.
- Maintains medicine and inventory records in the cloud.
- Provides secure and centralized data access.
- Supports updating and managing medicine information.



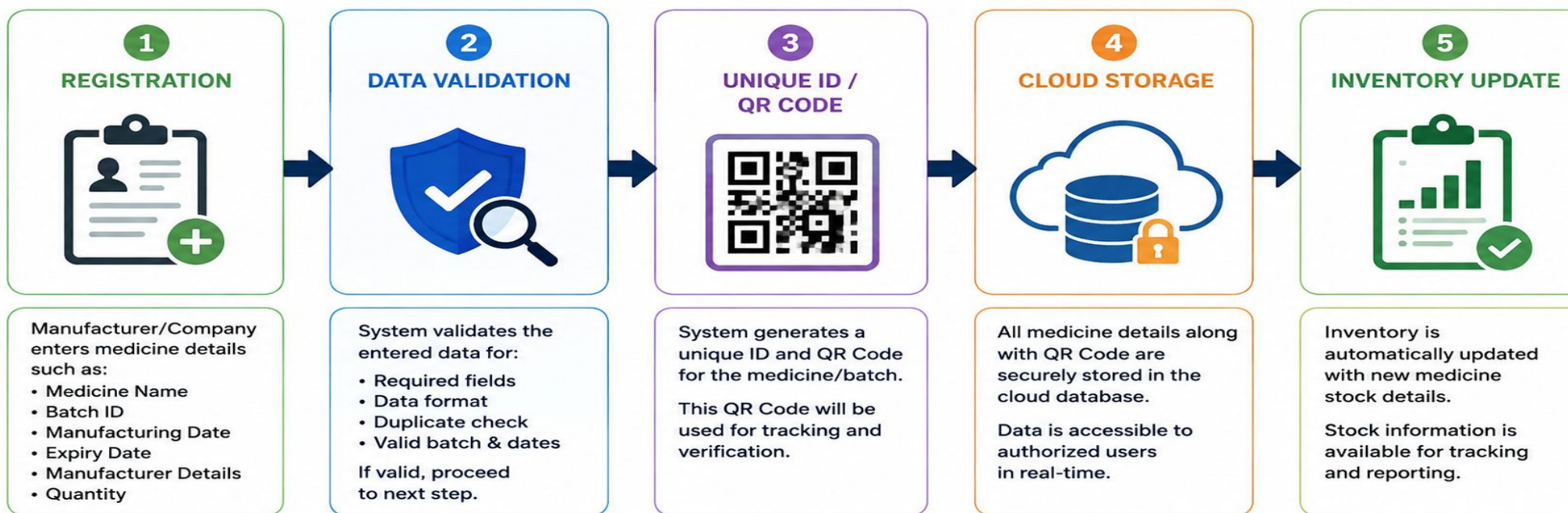
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WORKFLOW

MODULE 1 WORKFLOW

Registration → Data Validation → Unique ID/QR Code → Cloud Storage → Inventory Update



OUTCOME: Medicine is registered, verified, stored in cloud, and inventory is updated for secure tracking.

Module 2: Supply Chain Tracking

- Tracks medicine movement across the supply chain.
- Monitors **Manufacturer → Distributor → Wholesaler → Retailer → Consumer.**
- Records location and transaction details.
- Maintains complete movement history.
- Provides real-time status updates.
- Improves supply-chain visibility and accountability.

At every stage of the medicine's journey, the system records:

- **Batch ID** – Identifies the specific medicine batch.
- **Location** – Records where the medicine is currently located.
- **Date & Time** – Records when the medicine was received or transferred.
- **Current Status** – Shows whether it is dispatched, in transit, received, or delivered.
- **Transaction Details** – Records who sent and who received the medicine.

Module 3: Counterfeit Drug Detection

- Scans the medicine's **QR code/barcode**.
- Compares scanned information with cloud records.
- Verifies batch and manufacturer details.
- Checks expiry and product status.
- Identifies duplicate, fake, expired, or unauthorized medicines.
- Displays verification results to the user.

Detection Process

Scan QR/Barcode



Retrieve Cloud Data



Validate Medicine & Batch ID



Check Manufacturer + Expiry + Status



Verification Result

Result:



Genuine



Counterfeit



Expired



Unauthorized/Duplicate

Module 4: Monitoring Dashboard & Alerts

- Provides a centralized monitoring dashboard.
- Displays medicine stock and supply-chain status.
- Shows genuine, counterfeit, and expired medicines.
- Provides alerts for suspicious activities.
- Helps administrators monitor supply-chain performance.
- Supports report generation and data analysis.



Dashboard & Alert Workflow



Cloud Database



Data Processing & Analysis



Monitoring Dashboard



Status Visualization + Alerts



Admin / Manufacturer / Distributor

- **Alerts can include:**
- Counterfeit medicine detected
- Expired medicine detected
- Duplicate batch detected
- Unauthorized transaction
- Low stock notification



ADVANTAGES



- Real-time supply-chain visibility.
- Faster counterfeit detection.
- Centralized cloud-based data management.
- Better inventory monitoring.
- Improved transparency and traceability.
- Automated alerts and reporting.



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FUTURE SCOPE



- AI-based counterfeit prediction.
- IoT-based temperature monitoring.
- Blockchain-based medicine traceability.
- Mobile application for medicine verification.
- Advanced analytics for demand forecasting.

CONCLUSION

- The proposed system provides **end-to-end pharmaceutical supply-chain monitoring**.
- Cloud technology enables centralized and accessible medicine information.
- QR/barcode verification helps identify suspicious medicines.
- Real-time tracking improves transparency across the supply chain.
- Dashboards and alerts support faster decision-making.
- Overall, the system helps create a **safer, smarter, and more reliable pharmaceutical supply chain**.



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THANK YOU